

chirality interactions in cold atom optical lattices has been proposed³⁸.

Our model (8) is achieved at second order of the perturbation series. Higher order terms become truncation errors but may be controlled by small parameters $\lambda_{x,y,z}/J_{\text{cluster}} \sim \sqrt{|J_{x,y,z}|/J_{\text{cluster}}}$.

V. CONCLUSIONS.

We constructed the exactly solvable Kitaev honeycomb model¹ as the exact low energy effective Hamiltonian of a spin-1/2 model [equations (8) or (9)] with spin-rotation and time reversal symmetry. The spin in Kitaev model is represented as the pseudo-spin in the two-fold degenerate spin singlet subspace of a cluster of four antiferromagnetically coupled spin-1/2 moments. The physical spin model is a honeycomb lattice of such four-spin clusters, with certain inter-cluster interactions. The machinery for the exact mapping to pseudo-spin Hamiltonian was developed (see e.g. TABLE I), which is quite general and can be used to construct other interesting (exactly solvable) spin-1/2 models from spin rotation invariant systems.

In this construction the pseudo-spin correlations in the Kitaev model will be mapped to dimer or spin-chirality correlations in the physical spin system. The corresponding picture of the fractionalized Majorana fermion excitations and Ising vortices still remain to be clarified.

This exact construction contains high order physical spin interactions, which is undesirable for practical implementation. We described two possible approaches to reduce this problem: generating the high order spin interactions by perturbative expansion of the coupling to optical phonon, or the magnetic coupling between clusters. This perturbative construction will introduce truncation error of perturbation series, which may be controlled by small expansion parameters. Whether these constructions can be experimentally engineered is however beyond the scope of this study. It is conceivable that other perturbative expansion can also generate these high order spin interactions, but this possibility will be left for future works.

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Appendix A: Coupling between Distortions of a Tetrahedron and the Pseudo-spins

In this Appendix we reproduce from Ref.³⁵ the couplings of all tetrahedron distortion modes to the spin

system. And convert them to pseudo-spin notation in the physical spin singlet sector.

Consider a general small distortion of the tetrahedron, the spin Hamiltonian becomes

$$H_{\text{cluster, SL}} = (J_{\text{cluster}}/2) \left(\sum_{\ell} \mathbf{S}_{\ell} \right)^2 + J' \sum_{\ell < m} \delta r_{\ell m} (\mathbf{S}_{\ell} \cdot \mathbf{S}_m) \quad (\text{A1})$$

where $\delta r_{\ell m}$ is the change of bond length between spins ℓ and m , J' is the derivative of J_{cluster} with respect to bond length.

There are six orthogonal distortion modes of the tetrahedron [TABLE 1.1 in Ref.³⁵]. One of the modes A is the trivial representation of the tetrahedral group T_d ; two E modes form the two dimensional irreducible representation of T_d ; and three T_2 modes form the three dimensional irreducible representation. The E modes are also illustrated in FIG. 3.

The generic couplings in (A1) [second term] can be converted to couplings to these orthogonal modes,

$$J'(Q^A f^A + Q_1^E f_1^E + Q_2^E f_2^E + Q_1^{T_2} f_1^{T_2} + Q_2^{T_2} f_2^{T_2} + Q_3^{T_2} f_3^{T_2})$$

where Q are generalized coordinates of the corresponding modes, functions f can be read off from TABLE 1.2 of Ref.³⁵. For the A mode, $\delta r_{\ell m} = \sqrt{2/3} Q^A$, so f^A is

$$f^A = \sqrt{2/3} (\mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_3 \cdot \mathbf{S}_4 + \mathbf{S}_1 \cdot \mathbf{S}_3 + \mathbf{S}_2 \cdot \mathbf{S}_4 + \mathbf{S}_1 \cdot \mathbf{S}_4 + \mathbf{S}_2 \cdot \mathbf{S}_3).$$

The functions $f_{1,2}^E$ for the E modes have been given before but are reproduced here,

$$\begin{aligned} f_2^E &= (1/2)(\mathbf{S}_2 \cdot \mathbf{S}_4 + \mathbf{S}_1 \cdot \mathbf{S}_3 - \mathbf{S}_1 \cdot \mathbf{S}_4 - \mathbf{S}_2 \cdot \mathbf{S}_3), \\ f_1^E &= \sqrt{1/12}(\mathbf{S}_1 \cdot \mathbf{S}_4 + \mathbf{S}_2 \cdot \mathbf{S}_3 + \mathbf{S}_2 \cdot \mathbf{S}_4 + \mathbf{S}_1 \cdot \mathbf{S}_3 \\ &\quad - 2\mathbf{S}_1 \cdot \mathbf{S}_2 - 2\mathbf{S}_3 \cdot \mathbf{S}_4). \end{aligned}$$

The functions $f_{1,2,3}^{T_2}$ for the T_2 modes are

$$\begin{aligned} f_1^{T_2} &= (\mathbf{S}_2 \cdot \mathbf{S}_3 - \mathbf{S}_1 \cdot \mathbf{S}_4), \\ f_2^{T_2} &= (\mathbf{S}_1 \cdot \mathbf{S}_3 - \mathbf{S}_2 \cdot \mathbf{S}_4), \\ f_3^{T_2} &= (\mathbf{S}_1 \cdot \mathbf{S}_2 - \mathbf{S}_3 \cdot \mathbf{S}_4) \end{aligned}$$

Now we can use TABLE I to convert the above couplings into pseudo-spin. It is easy to see that f^A and $f_{1,2,3}^{T_2}$ are all zero when converted to pseudo-spins, namely projected to the physical spin singlet sector. But $f_1^E = (P_{14} + P_{23} + P_{24} + P_{13} - 2P_{12} - 2P_{34})/(4\sqrt{3}) = -(\sqrt{3}/2)\tau^x$ and $f_2^E = (P_{24} + P_{13} - P_{14} - P_{23})/4 = (\sqrt{3}/2)\tau^y$. This has already been noted by Tchernyshyov *et al.*²⁸, only the E modes can lift the degeneracy of the physical spin singlet ground states of the tetrahedron. Therefore the general spin lattice coupling is the form of (12) given in the main text.